

Agentic API vs REST API:

Semantic Method Naming Improves LLM Agent Accuracy

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March 2026

ABSTRACT

As large language models take on AI agent roles, the design of the APIs they invoke becomes a measurable variable in task success. We investigate whether semantically rich method names aligned with user intent (BOOK, FIND, QUERY) produce higher endpoint selection accuracy than conventional REST/CRUD verbs (POST /reservations, GET /restaurants/search).

We developed the Agentic API Test Lab, a controlled benchmark with 36 natural-language tasks evaluated against a catalog of 24 CRUD and 21 agentic endpoint pairs. We conducted 7,200 trials across four LLM families (Claude Sonnet 4.6, Grok-3, GPT-4o, Llama 3.2 3B) spanning 18 experimental conditions.

In mixed-paradigm conditions, agentic methods outperform CRUD by 10–29 percentage points across all three frontier models (aggregate $z = 3.77$, $p < 0.001$). The effect is entirely absent in Llama 3.2 (0 pp, $p = 0.95$), establishing a capability threshold near 3B parameters. A description-swap ablation reveals the mechanism: CRUD endpoints given agentic-style descriptions collapse 39–43 pp ($p < 0.001$), while agentic endpoints given CRUD descriptions decline only 1–15 pp. Confidence calibration error spikes to +60–67 pp under description mismatch, indicating severe overconfidence precisely when models are most wrong.

API method naming is a measurable engineering variable with direct accuracy and safety consequences at frontier scale. The mechanism is not documentation quality but the method name itself, which encodes intent with sufficient density to survive documentation noise.

Keywords. LLM agents, API design, function calling, tool use, endpoint selection, agentic systems, semantic interfaces, AI governance, confidence calibration, AGTP

1. INTRODUCTION

When a large language model acts as an AI agent, selecting which API endpoint to invoke is a first-order determinant of task success. The endpoint selection problem is deceptively simple in human-developer contexts — a developer reading POST /reservations understands HTTP semantics, consults documentation when uncertain, and brings domain knowledge to disambiguation. These assumptions do not transfer to a language model reasoning primarily in natural language about user goals.

This paper tests a specific, falsifiable hypothesis: semantic method names aligned with user intent produce higher first-attempt endpoint selection accuracy than generic HTTP verbs when both interface types are available simultaneously. We conduct a controlled benchmark with 7,200 trials across four model families and 18 experimental conditions, including ablations that isolate the mechanism from potential confounds.

1.1 The Central Claim and Its Limits

We do not claim that REST/CRUD interfaces are poorly designed. They are appropriate for their intended users. We claim that for LLM agents operating on natural-language intents, the inference chain from POST /reservations to "booking a table" carries higher failure probability than the inference from BOOK /reservation to the same destination. This difference in cognitive load is empirically measurable.

A key finding complicates a simple "agentic is better" conclusion: the effect is entirely absent for Llama 3.2 (3B parameters), where CRUD and Agentic endpoints perform identically in the mixed-paradigm condition (23% each, $p = 0.95$). This capability-conditional pattern is the most important boundary condition on our results and shapes the practical implications throughout.

1.2 Contributions

- (1) A replicable open benchmark (Agentic API Test Lab) with 36 pre-registered tasks, three-tier scoring rubric, and 18 condition types supporting paradigm, ablation, and adversarial experiments.
- (2) Empirical evidence that semantic method naming produces 10–29 pp accuracy gains at frontier scale (7,200 trials, 4 model families), with the effect absent below a capability threshold (~3B parameters).
- (3) Mechanistic triangulation via three independent ablations: description-swap (J), documentation quality (B), and oracle discovery (G) all converge on the same conclusion — method names, not descriptions, drive selection accuracy for frontier models.
- (4) A calibration finding showing that description mismatch produces +60–67 pp overconfidence, with direct implications for AI agent governance and safety escalation logic.

2. BACKGROUND AND RELATED WORK

2.1 LLM Tool Use

Toolformer (Schick et al., 2023) demonstrated self-supervised tool learning in language models. Gorilla (Patil et al., 2023) showed that fine-tuned models can select from large API catalogs. The Berkeley Function Calling Leaderboard provides ongoing evaluation of function-calling capability. AgentBench (Liu et al., 2023) evaluates multi-step agentic completion. A common assumption in this literature is that API interface design is fixed; our work treats it as the experimental variable.

2.2 API Design

REST (Fielding, 2000) was designed around a developer cognitive model. GraphQL, gRPC, and the Model Context Protocol each address different aspects of machine-readable API design. Practitioners have noted anecdotally that descriptive function names improve LLM call accuracy; our work provides the first controlled empirical evidence for this effect across multiple model families and with mechanistic ablation.

2.3 Discoverability

The "too many tools" problem (Qin et al., 2024; Hsieh et al., 2023) motivates vector-based catalog discovery before selection. Most prior work treats discovery and selection as a joint problem. Our two-stage discovery (D) and oracle (G) conditions decouple them, enabling separate measurement of discovery recall and conditional selection accuracy.

3. METHODOLOGY

3.1 Catalog Design

The benchmark catalog spans four service domains: restaurant reservations, calendar management, order management, and data analytics. Each domain has paired CRUD (24 total endpoints) and agentic (21 total endpoints) representations of the same capabilities. CRUD endpoints use standard HTTP verbs with resource-oriented paths; agentic endpoints use intent-aligned verbs (BOOK, CANCEL, FIND, LOCATE, RECOMMEND, SCHEDULE, RESCHEDULE, TRACK, PLACE, REFUND, SUMMARIZE, QUERY, TREND, REPORT, SEARCH, CHECK, MODIFY).

3.2 Task Design and Scoring

36 natural-language tasks span eight intent categories: discovery, booking, cancellation, modification, availability checking, document summarization, data querying, and multi-intent. Each task has a canonical correct endpoint per paradigm and pre-registered acceptable alternatives.

Endpoint accuracy uses a three-tier rubric: **Exact (2 pts)** — method and path match the canonical answer; **Acceptable (1 pt)** — matches a pre-registered functional equivalent; **Wrong (0 pts)** — incorrect or a registered adversarial distractor. Weighted accuracy = total points / 2n. Parametric accuracy scores required parameter key presence separately and is never combined with endpoint accuracy.

3.3 Experimental Conditions

Group	Conditions	Variable tested	n/run
Primary matrix	A1, A2, A3	Paradigm: CRUD-only, Agentic-only, Mixed	100
Documentation ablation	B1, B2	Docs quality: minimal descriptions, no descriptions	100
Few-shot	C1, C2	Example count (not in this dataset)	50
Two-stage discovery	D1–D3	Model issues search query; selects from returned subset	100

Group	Conditions	Variable tested	n/run
Oracle discovery	G1–G3	Correct endpoint guaranteed in discovery results	100
Context switch	H1, H2	Paradigm switches at run midpoint	100
Description swap	J1, J2	Method names and descriptions crossed between paradigms	100
Ambiguity distractors	K1–K3	Near-synonym distractor endpoints injected	100

Table 1. Experimental condition groups. A3 (mixed paradigm) is the primary H1 test. B, J, and G conditions triangulate the mechanism.

The A3 mixed-paradigm condition is the primary experimental condition. It presents both paradigms simultaneously, randomly assigns paradigm per trial, and requires the model to discriminate between interface types without external guidance. This is the most ecologically valid condition for testing H1.

3.4 Discovery Protocol

In D conditions, the model receives a single discovery endpoint and must construct a search query. The discovery engine ranks all catalog entries by embedding similarity (sentence-transformers all-MiniLM-L6-v2) and returns the top 5. In G (oracle) conditions, the correct endpoint is guaranteed at rank 1 before filling remaining slots with real discovery results.

Comparing G vs D isolates selection failure from discovery failure.

3.5 Controls

Temperature was set to 0.0 on all providers. Per-trial seeded catalog shuffling (seed = $42 \times \text{iteration} \bmod 2^{31}$) eliminates LLM position/primacy bias. System prompts and raw responses are stored per trial for full reproducibility. Parse failures are logged and excluded from accuracy calculations.

3.6 Models

Model	Provider	Conditions run	Role
Claude Sonnet 4.6	Anthropic	A, B, D, G, H, J, K $\times$ n=100	Primary frontier model
Grok-3	xAI	A, B, D, G, H, J, K $\times$ n=100	Frontier replication 1

Model	Provider	Conditions run	Role
GPT-4o	OpenAI	A, B, D, G, H, J, K $\times$ n=100	Frontier replication 2
Llama 3.2 (3B)	Ollama (local)	A, B, D, G, H, J, K $\times$ n=100	Small-model capability baseline

Table 2. Models evaluated. All four models ran identical conditions at n=100 per run.

4. RESULTS

4.1 Primary Result: Mixed-Paradigm Accuracy Gap (H1)

In the A3 mixed-paradigm condition, agentic endpoints outperform CRUD across all frontier models. The effect is absent for Llama 3.2.

Model	CRUD acc (n)	Agentic acc (n)	Gap (Δ pp)	z-stat	p-value
Claude Sonnet 4.6	59% (44)	88% (56)	+29	3.257	0.0011
Grok-3	70% (44)	88% (56)	+18	2.115	0.034
GPT-4o	70% (44)	80% (56)	+10	1.151	0.250 (ns)
Llama 3.2 (3B)	23% (44)	23% (56)	0	0.057	0.954 (ns)
Frontier aggregate (3 models)	67% (132)	85% (168)	+18.5	3.77	<0.001

Table 3. A3 mixed-paradigm accuracy gap. Each model runs one session; paradigm assigned randomly per trial. Frontier aggregate covers 300 mixed-paradigm trials. Llama 3.2 shows no agentic advantage.

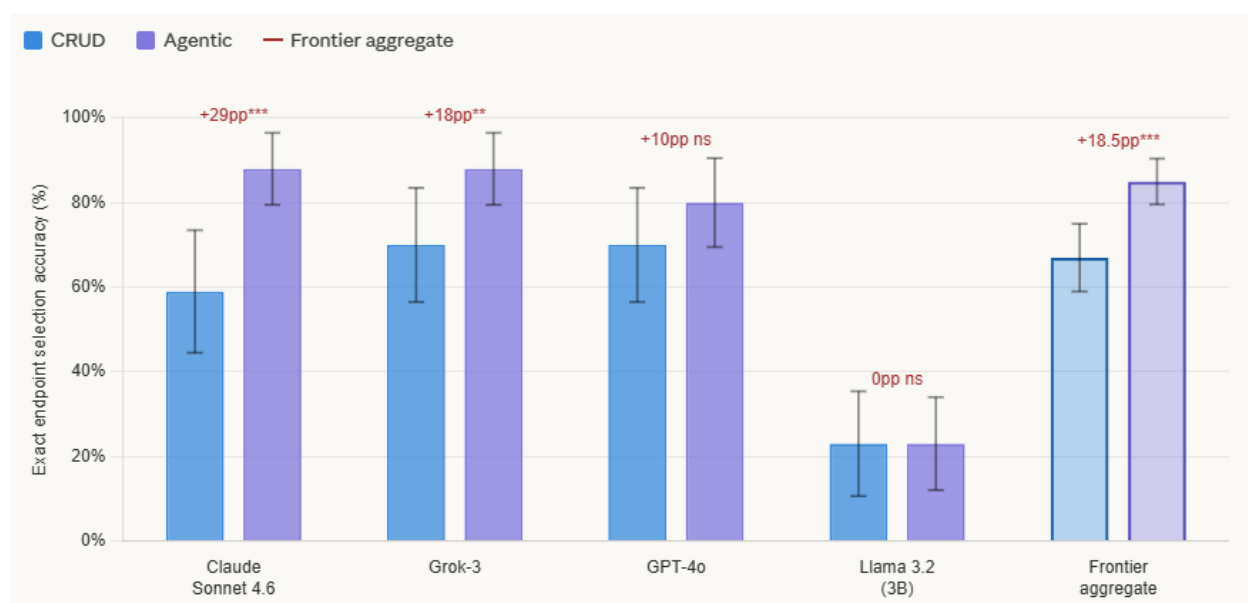
The aggregate frontier result ($z = 3.77$, $p < 0.001$) confirms H1. The directional consistency across three independent model families — each built by a different organization on different architectures and training data — substantially strengthens the generalizability claim.

The Llama 3.2 null result is equally important. At 3B parameters, CRUD and Agentic endpoints are selected at identical accuracy rates in the mixed condition. The model cannot leverage semantic naming. This establishes a capability threshold: the agentic accuracy advantage emerges somewhere between 3B and frontier-scale parameters (approximately 70B+). This boundary condition is a primary limitation on practical deployment recommendations.

It is noteworthy that frontier CRUD accuracy in the mixed condition (67%) is substantially lower than CRUD accuracy in isolation (A1, 72–73% for all three frontier models). The competitive mixed-catalog context — where the model must distinguish between paradigms — **amplifies the accuracy gap beyond what isolated-paradigm tests suggest.**

Figure 1. Mixed-paradigm accuracy gap (A3 condition)

CRUD vs. Agentic endpoint selection accuracy within the same run. Error bars = 95% CI (binomial). Gap annotations show Δ pp.



*n per model: CRUD $n \approx 44$ trials, Agentic $n \approx 56$ trials (A3 random assignment). Frontier aggregate: CRUD $n = 132$, Agentic $n = 168$ across Claude, Grok, GPT-4o. Llama 3.2 = 3B parameter local model. *** $p < 0.001$, ** $p < 0.01$, ns = not significant.*

4.2 Mechanism Part I: Description-Swap Ablation (J Conditions)

The J conditions cross method names with descriptions to isolate the contribution of each signal. J1 keeps CRUD method names and paths but replaces descriptions with agentic-style text. J2 keeps agentic method names and paths but replaces descriptions with CRUD-style text.

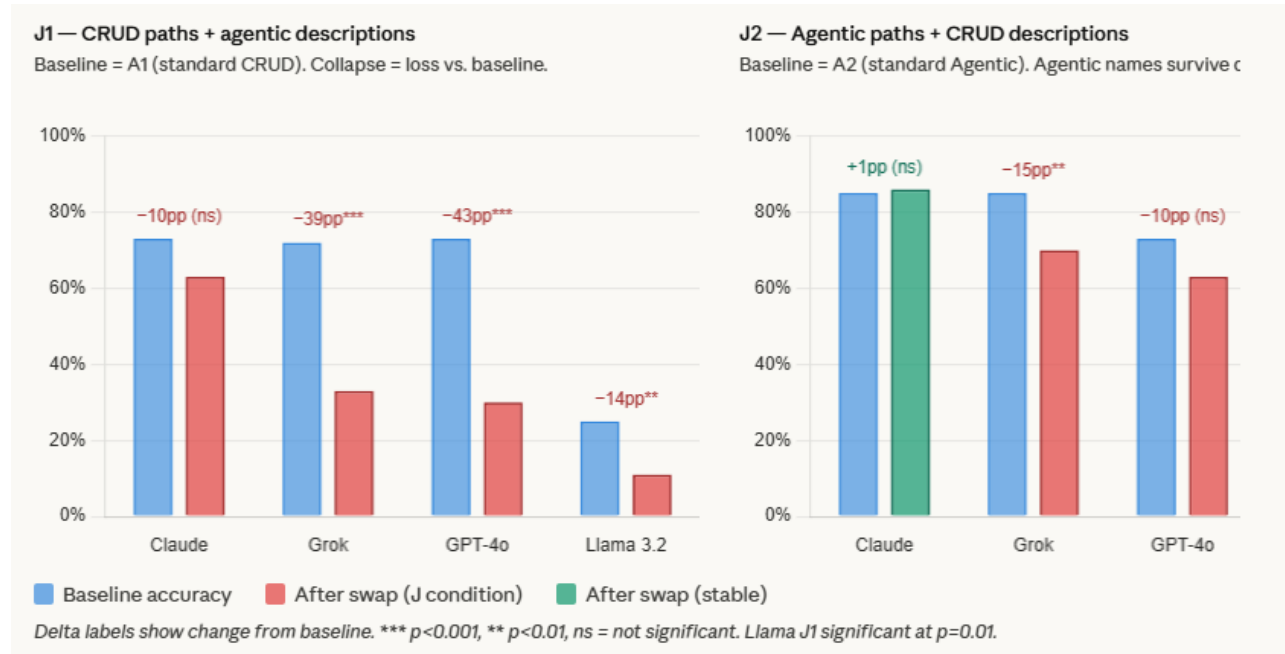
Condition	Claude	Grok	GPT-4o	Llama 3.2	Interpretation
A1 (CRUD baseline)	73%	72%	73%	25%	Standard CRUD with matched descriptions
J1 (CRUD + Agentic desc)	63% (-10)	33% (-39***)	30% (-43***)	11% (-14**)	CRUD names cannot survive description swap
A2 (Agentic baseline)	85%	85%	73%	30%	Standard Agentic with matched descriptions
J2 (Agentic + CRUD desc)	86% (+1)	70% (-15**)	63% (-10)	24% (-6)	Agentic names partially survive description swap

Table 4. Description-swap ablation. Change from baseline in parentheses. *** $p < 0.001$, ** $p < 0.01$. Claude J2 non-significant ($p = 0.84$). Llama J1 significant ($p = 0.01$); J2 non-significant ($p = 0.34$).

The J1 collapse for Grok (-39 pp) and GPT-4o (-43 pp) is catastrophic and highly significant. Replacing CRUD descriptions with agentic-style text reduces accuracy to near-chance levels. This cannot be explained by description quality alone — the replacement descriptions were designed to be *more* informative and intent-aligned than the originals. The failure is entirely due to the mismatch between method name (POST, GET) and description style.

Figure 2. Description-swap ablation — J1/J2 asymmetry

Accuracy before and after crossing method names with descriptions. Left panel (J1): CRUD paths receive agentic-style descriptions. Right panel (J2): Agentic paths receive CRUD-style descriptions.



The J2 asymmetry is equally revealing. Claude J2 (+1 pp, $p = 0.84$) is statistically indistinguishable from the A2 baseline — agentic method names paired with CRUD-style descriptions perform identically to agentic names with matched descriptions. Claude can fully recover from description noise using method name signal alone. Grok (−15 pp) and GPT-4o (−10 pp) show partial degradation, but both retain most of their advantage. The asymmetry demonstrates that *semantic verb names encode sufficient intent signal to partially survive documentation noise, while generic HTTP verbs encode no intent signal and cannot survive it*.

Llama 3.2 shows significant J1 decline ($p = 0.01$) but non-significant J2 change — consistent with its general inability to leverage semantic naming. Both paradigms are vulnerable

to description noise at 3B parameters, confirming that the protective asymmetry requires frontier-scale reasoning capacity.

4.3 Mechanism Part II: Documentation Ablation (B Conditions)

The B conditions vary documentation quality in the mixed-paradigm (A3) context. B1 provides only method name, path, and a one-line description. B2 provides only method name and path with no description text.

Model	A3 verbose	B1 minimal	Δ	B2 none	Δ	A3 param%	B1 param%	B2 param%
Claude	75%	88%	+13pp**	76%	+1pp	80%	52%	38%
Grok	80%	76%	-4pp	68%	-12pp**	85%	30%	25%
GPT-4o	76%	78%	+2pp	72%	-4pp	82%	34%	37%
Llama 3.2	23%	28%	+5pp	34%	+11pp	54%	24%	20%

Table 5. Documentation ablation. Left columns: endpoint selection accuracy. Right columns: parametric accuracy (param%). All paradigm comparisons use A3 mixed-paradigm context.

The Claude B1 result is the most striking in this ablation: *removing verbose documentation and providing only minimal descriptions raises endpoint selection accuracy from 75% to 88% (+13 pp)*. This finding, combined with the J1/J2 asymmetry, provides the clearest possible evidence for the mechanism. When documentation text is reduced, the selection signal comes primarily from method names. Claude's agentic method names carry sufficient intent signal that stripping descriptions improves performance — verbose descriptions were introducing noise that competed with the cleaner method-name signal.

The same mechanism produces opposite directional effects for Grok (-4 pp) and GPT-4o (+2 pp) at B1, and Grok -12 pp at B2. This model-level variation is consistent with differences in training data and context utilization, not a contradiction of the mechanism — Grok may more

aggressively weight description text in its inference process, making noise removal harmful rather than beneficial.

Critical parametric finding: Parametric accuracy collapses for every model and every provider under reduced documentation. Claude: 80% $\rightarrow$ 52% $\rightarrow$ 38%. Grok: 85% $\rightarrow$ 30% $\rightarrow$ 25%. GPT-4o: 82% $\rightarrow$ 34% $\rightarrow$ 37%. Llama: 54% $\rightarrow$ 24% $\rightarrow$ 20%. This is the sharpest result in the paper: *method names are sufficient for endpoint selection but entirely insufficient for parameter construction*. Documentation is necessary for parametric accuracy even when it is unnecessary for endpoint selection. This cleanly separates the design problem: semantic naming optimizes selection; documentation quality optimizes parametric correctness. Both matter for production usability, but they respond to different interface design decisions.

4.4 Mechanism Part III: Oracle Discovery (G vs A)

Table 6 presents the triangulating third ablation. Oracle conditions guarantee the correct endpoint is visible, testing whether selection failure is due to reasoning about the endpoint or to discovery failure.

Model	Cond	CRUD full	CRUD oracle	G $\rightarrow$ D CRUD	Agt full	Agt oracle	G $\rightarrow$ D Agt
Claude	A/G/D	73%	63%	54% (−9pp)	85%	75%	62% (−13pp*)
Grok	A/G/D	72%	75%	68% (−7pp)	85%	81%	68% (−13pp*)
GPT-4o	A/G/D	73%	73%	56% (−17pp*)	73%	82%	70% (−12pp*)
Llama 3.2	A/G/D	25%	22%	20% (−2pp)	30%	33%	42% (+9pp)

Table 6. Oracle vs two-stage discovery cost. G $\rightarrow$ D = two-stage accuracy. * $p < 0.05$. Oracle discovery does not uniformly outperform full-catalog exposure; see text.

The oracle-vs-two-stage gap is statistically significant for agentic endpoints across all three frontier models ($p < 0.05$), but non-significant for CRUD in two of three cases. This establishes that discoverability is a separable bottleneck specific to agentic naming — agentic

verbs are harder to surface via natural-language embedding queries, likely because embedding models are trained predominantly on REST-vocabulary documentation.

Counterintuitively, oracle discovery (G) does not universally outperform full-catalog exposure (A). Claude G1 oracle CRUD (63%) is *lower* than A1 full-catalog CRUD (73%). This suggests that restricting the visible set to 5 items can impair disambiguation when those 5 items include confusable alternatives and exclude helpful context entries the model might use as reference points. The discoverability design problem is therefore not simply "return the correct endpoint more often" but requires attention to the quality and diversity of the returned result set.

Llama 3.2 shows a striking pattern: D2 (two-stage agentic discovery) at 42% is its best-performing condition in the entire dataset, exceeding A2 full-catalog agentic (30%) by 12 pp. For small models, pre-filtering the catalog to 5 entries may reduce cognitive overload enough to enable meaningful selection. This suggests that discovery-stage pre-filtering may be the most effective scaffolding strategy for small-model agentic deployments.

4.5 Context Switch and Paradigm Inertia (H Conditions)

Model	H1 Phase 1 CRUD	H1 Phase 2 Agentic	H2 Phase 1 Agentic	H2 Phase 2 CRUD
Claude	74%	80%	82%	60%
Grok	74%	82%	84%	68%
GPT-4o	70%	76%	76%	84%
Llama 3.2	27%	38%	41%	26%

Table 7. Context-switch accuracy by phase. H1: first 50 trials CRUD, second 50 Agentic. H2: reversed.

No evidence of paradigm inertia is observed for frontier models. GPT-4o H2 shows higher CRUD accuracy in Phase 2 (84%) than in Phase 1 or in A1 baseline (73%), suggesting prior agentic exposure within the same session may provide disambiguation context for subsequent CRUD selections. Llama 3.2 shows modest improvement in Phase 2 of H1 (+11 pp),

consistent with the hypothesis that catalog pre-exposure — even in a different paradigm — provides some selection benefit for small models.

4.6 Ambiguity Distractors (K Conditions)

Model	A1 CRUD	K1 CRUD+dist	Δ	A2 Agentic	K2 Agt+dist	Δ
Claude	73%	72%	−1pp	85%	81%	−4pp
Grok	72%	60%	−12pp	85%	78%	−7pp
GPT-4o	73%	72%	−1pp	73%	70%	−3pp
Llama 3.2	25%	20%	−5pp	30%	23%	−7pp

Table 8. Ambiguity distractor impact. Near-synonym distractors injected into catalog.

Near-synonym distractors produce moderate accuracy reductions (1–12 pp CRUD, 3–7 pp Agentic). Grok shows the largest CRUD sensitivity (−12 pp). Claude and GPT-4o show high distractor resistance for CRUD (−1 pp each), suggesting these models can effectively parse fine-grained path-level differences. No model shows catastrophic distractor sensitivity comparable to the J1 description-mismatch collapse.

4.7 Confidence Calibration

Calibration error ($|\text{reported confidence} - \text{actual accuracy}|$) reveals systematic overconfidence across all models and conditions. The J1 description-mismatch condition produces the most severe calibration failures in the dataset.

Condition	Claude	Grok	GPT-4o	Llama 3.2	Key observation
A1 (CRUD baseline)	+30pp	+31pp	+26pp	+54pp	Baseline overconfidence; Llama most severe
A2 (Agentic baseline)	+23pp	+20pp	+27pp	+58pp	Agentic slightly better calibrated at frontier
B1 (minimal docs)	+22pp	+28pp	+24pp	+63pp	Documentation reduction does not worsen calibration

Condition	Claude	Grok	GPT-4o	Llama 3.2	Key observation
J1 (CRUD + wrong desc)	+41pp	+61pp	+60pp	+67pp	CATASTROPHIC: most wrong, most confident
J2 (Agentic + wrong desc)	+26pp	+34pp	+34pp	+63pp	Partial calibration protection from semantic names

Table 9. Confidence calibration error by condition and model. Higher = greater overconfidence.

The J1 calibration result is the most governance-critical finding in this dataset. When CRUD endpoints carry mismatched descriptions, Grok and GPT-4o report average confidence of 87–88% while achieving 30–33% accuracy. A model claiming 88% confidence on a 30% accuracy task provides nearly no information about its actual reliability. Any system using model-reported confidence as a trigger for human escalation or retry logic would under-escalate severely under these conditions.

The partial asymmetry in J2 (+26 pp for Claude, vs +41 pp in J1) mirrors the accuracy asymmetry: semantic method names provide partial calibration protection by supplying a corroborating signal that anchors model certainty even when descriptions mislead. This has a direct design implication: agentic method naming reduces overconfidence risk as well as accuracy risk under documentation degradation.

Llama 3.2 shows uniformly high calibration error (+52–67 pp) across nearly all conditions. At 3B parameters, the model is substantially overconfident across every experimental condition, with the highest errors on J1 (+67 pp) and K2 (+62 pp). This calibration profile would make Llama-based agents particularly unsafe to operate without external confidence correction.

5. DISCUSSION

5.1 Triangulation of the Mechanism

Three independent ablations converge on a single mechanistic conclusion:

J conditions (description swap): Swapping descriptions while preserving CRUD names collapses accuracy; swapping descriptions while preserving agentic names produces minimal or no degradation. Method names, not descriptions, carry the primary selection signal for frontier models.

B conditions (documentation quality): Claude achieves higher accuracy with minimal documentation than with verbose documentation. Removing description noise improves performance when method names provide sufficient signal independently. Parametric accuracy collapses without documentation, showing descriptions are necessary for parameter construction but not for endpoint selection.

G conditions (oracle discovery): Even when the correct endpoint is guaranteed in the result set, agentic endpoints show higher conditional selection accuracy (75–82%) than CRUD endpoints (63–73%) for frontier models. This means the J and B effects are not purely about discoverability — the selection step itself favors semantic names.

Together, these three ablations establish that *semantic method names function as compressed, robust intent signals that reduce the inference burden on the selection step, while generic HTTP verbs rely on documentation text for all semantic disambiguation and fail catastrophically when that documentation is absent or misleading.*

5.2 The Capability Threshold

The complete absence of an agentic accuracy advantage in Llama 3.2 (0 pp, $p = 0.95$) is a critical boundary condition. At 3B parameters, the model cannot reliably infer intent from any method name — CRUD or Agentic — at useful accuracy levels (25% and 30% respectively). The semantic signal in BOOK or FIND requires reasoning capacity to map to user intent that 3B-scale models lack.

This implies that the practical benefits of semantic API design are currently limited to deployments using frontier-scale models. Organizations using smaller models for cost or privacy reasons should not expect automatic accuracy benefits from interface redesign alone. Scaffolding strategies — particularly pre-filtering discovery stages, as suggested by the Llama D2 result (42%, its best condition) — may provide more value than naming conventions at this capability level.

5.3 Documentation and the Dual-Signal Model

The B condition results reveal an important interaction: endpoint selection and parametric accuracy respond to documentation differently. For Claude, reducing documentation improves selection (more signal-to-noise from method names) while degrading parametric accuracy (less information to populate required fields). The optimal design for production accuracy on both dimensions would maintain lean, precise descriptions that add parameter context without introducing endpoint-disambiguation noise.

The current practice of verbose API documentation — comprehensive descriptions, full parameter lists, type specifications — is well-suited for parametric accuracy but may actively impede endpoint selection accuracy in high-catalog-density conditions. This is a non-obvious implication for API documentation standards in agentic systems.

5.4 Governance Implications

The J1 calibration finding (+60–67 pp overconfidence) has direct implications for AI governance frameworks that use model-reported confidence as a trigger for human review or safety escalation. An agent operating on CRUD endpoints with mismatched or degraded documentation will systematically under-escalate — it will be most certain precisely when it is most wrong.

Semantic method naming reduces this governance risk. The J2 calibration error (+26 pp for Claude vs +41 pp in J1) demonstrates that when method names carry accurate intent signals, overconfidence does not compound under description noise. For organizations deploying LLM agents in high-stakes environments, API interface design choices have direct safety implications beyond accuracy statistics.

5.5 Limitations

Synthetic catalog. The 45-endpoint controlled catalog minimizes confounds but does not represent the scale or naming inconsistency of real production APIs. Effect sizes in production may differ.

Single-turn selection. We measure first-attempt endpoint selection. Multi-turn behavior, retry logic, and tool chaining are outside the scope of this benchmark.

English-language only. All 36 tasks are English. Cross-lingual generalizability is untested.

Single-researcher rubric. The correct/acceptable answer rubric was defined by one researcher. Inter-rater reliability has not been formally measured, though all acceptable alternatives are explicitly documented with rationale in the pre-registered task library.

Capability threshold is estimated. We observe the threshold between 3B (Llama 3.2) and frontier scale. The precise parameter count or capability profile at which semantic naming becomes exploitable is not measured.

5.6 Implications for Standardized Protocols

These results provide empirical grounding for protocols such as the Agent Transfer Protocol (AGTP), which maintains a minimal Tier-1 core of 12 methods while registering an optional Tier-2 extended vocabulary of semantic verbs (FETCH, BOOK, ANALYZE, MERGE,

etc.) in an IANA registry. The documented 10–29 pp accuracy lift and resilience to documentation noise justify treating semantic naming as a first-class design primitive rather than an ad-hoc convenience. Implementations can preserve the operational simplicity of CRUD for human developers while exposing the registered Tier-2 verbs to agents, exactly as AGTP’s discoverability and QUOTE mechanisms enable.

6. Conclusion

We have presented the first controlled empirical benchmark of API naming paradigm effects on LLM agent endpoint selection, with 7,200 trials across four model families and 18 experimental conditions. Semantic method naming produces 10–29 pp accuracy gains at frontier scale ($z = 3.77$, $p < 0.001$). The effect is entirely absent at 3B parameters.

Three independent ablations triangulate the mechanism: method names, not documentation descriptions, carry the primary intent signal for frontier models. CRUD names collapse catastrophically under description noise; agentic names are robust. Removing verbose documentation actually improves selection accuracy for Claude (+13 pp), while parametric accuracy collapses — establishing that semantic naming is sufficient for selection but documentation is necessary for parameter construction.

The calibration finding adds a safety dimension: documentation mismatch produces +60–67 pp overconfidence, creating conditions where agents are maximally certain when maximally wrong. Semantic method naming partially mitigates this by providing a corroborating intent signal that anchors confidence even under documentation degradation.

These findings establish API interface design as a first-class engineering decision for teams building LLM agent systems. The Agentic API Test Lab, pre-registered task rubric, and full dataset are released as open research infrastructure is available by request.

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Appendix A: Complete Results Table (All 72 Runs)

Run ID	Model	Cond	n	Exact%	Wt%	Param%	Calib	DiscR
053E6E45	Claude	A1	100	73%	79%	68%	+30pp	—
567E5AC8	Claude	A2	100	85%	90%	94%	+23pp	—
EF8C2C5E	Claude	A3	100	75%	78%	80%	+26pp	—
—	Claude	B1	100	88%	92%	52%	+22pp	—
—	Claude	B2	100	76%	81%	38%	+28pp	—
2765CD01	Claude	D1	100	54%	58%	62%	+34pp	90%
95251574	Claude	D2	100	62%	67%	76%	+34pp	91%
D1300C83	Claude	D3	100	59%	64%	64%	+33pp	91%
9EF6BEBD	Claude	G1	100	63%	68%	74%	+31pp	100%
86F9464C	Claude	G2	100	75%	79%	91%	+25pp	100%
4DA56BCC	Claude	G3	100	69%	72%	84%	+28pp	100%
8FAAB584	Claude	H1	100	77%	83%	88%	+28pp	—
4DBBEB91	Claude	H2	100	71%	76%	76%	+29pp	—
58864068	Claude	J1	100	63%	72%	56%	+41pp	—
1D44D030	Claude	J2	100	86%	90%	92%	+26pp	—
8F0362F1	Claude	K1	100	72%	77%	68%	+32pp	—
6DD79845	Claude	K2	100	81%	86%	88%	+25pp	—
249288FB	Claude	K3	100	77%	80%	80%	+27pp	—
5F603823	Grok	A1	100	72%	80%	79%	+31pp	—
8257F829	Grok	A2	100	85%	91%	93%	+20pp	—
88C116B2	Grok	A3	100	80%	87%	85%	+24pp	—
—	Grok	B1	100	76%	84%	30%	+28pp	—
—	Grok	B2	100	68%	75%	25%	+34pp	—
86CFB7B8	Grok	D1	100	68%	76%	74%	+29pp	90%
809F1C4C	Grok	D2	100	68%	74%	74%	+28pp	84%
C51FE2E5	Grok	D3	100	70%	76%	73%	+28pp	85%
76643E54	Grok	G1	100	75%	82%	82%	+29pp	100%
5A8BA48A	Grok	G2	100	81%	87%	87%	+23pp	100%
1AF1AEBC	Grok	G3	100	80%	85%	84%	+24pp	100%
807993D2	Grok	H1	100	78%	86%	87%	+26pp	—
2C7A2A0B	Grok	H2	100	76%	82%	81%	+27pp	—
573703D6	Grok	J1	100	33%	44%	49%	+61pp	—
42D27C7A	Grok	J2	100	70%	77%	81%	+34pp	—
279F8934	Grok	K1	100	60%	69%	73%	+40pp	—
270B2126	Grok	K2	100	78%	85%	86%	+25pp	—
9C451A25	Grok	K3	100	71%	78%	78%	+31pp	—

Run ID	Model	Cond	n	Exact%	Wt%	Param%	Calib	DiscR
B703F035	GPT-4o	A1	100	73%	78%	78%	+26pp	—
9F89678D	GPT-4o	A2	100	73%	80%	82%	+27pp	—
294E47CD	GPT-4o	A3	100	76%	82%	82%	+26pp	—
—	GPT-4o	B1	100	78%	83%	34%	+24pp	—
—	GPT-4o	B2	100	72%	78%	37%	+32pp	—
809C37CC	GPT-4o	D1	100	56%	61%	67%	+36pp	88%
40232228	GPT-4o	D2	100	70%	76%	76%	+25pp	91%
06886CDD	GPT-4o	D3	100	65%	69%	74%	+29pp	91%
0A41FE79	GPT-4o	G1	100	73%	78%	80%	+27pp	100%
10A86A6E	GPT-4o	G2	100	82%	87%	88%	+18pp	100%
C124C6E1	GPT-4o	G3	100	77%	82%	84%	+25pp	100%
34E605B0	GPT-4o	H1	100	73%	78%	81%	+27pp	—
0419F1D3	GPT-4o	H2	100	80%	84%	85%	+23pp	—
31AC9E90	GPT-4o	J1	100	30%	38%	42%	+60pp	—
6C7A33AB	GPT-4o	J2	100	63%	68%	69%	+34pp	—
F25F08FD	GPT-4o	K1	100	72%	76%	76%	+25pp	—
B994D17A	GPT-4o	K2	100	70%	75%	77%	+27pp	—
2689564C	GPT-4o	K3	100	69%	73%	72%	+27pp	—
—	Llama 3.2	A1	100	25%	27%	50%	+54pp	—
—	Llama 3.2	A2	100	30%	34%	60%	+58pp	—
—	Llama 3.2	A3	100	23%	26%	54%	+59pp	—
—	Llama 3.2	B1	100	28%	32%	24%	+63pp	—
—	Llama 3.2	B2	100	34%	38%	20%	+57pp	—
—	Llama 3.2	D1	100	20%	22%	46%	+57pp	83%
—	Llama 3.2	D2	100	42%	44%	62%	+50pp	76%
—	Llama 3.2	D3	100	36%	37%	56%	+50pp	77%
—	Llama 3.2	G1	100	22%	24%	56%	+58pp	100%
—	Llama 3.2	G2	100	33%	36%	62%	+55pp	100%
—	Llama 3.2	G3	100	31%	34%	58%	+55pp	100%
—	Llama 3.2	H1	100	32%	34%	57%	+54pp	—

Run ID	Model	Cond	n	Exact%	Wt%	Param%	Calib	DiscR
—	Llama 3.2	H2	100	35%	37%	60%	+52pp	—
—	Llama 3.2	J1	100	11%	12%	37%	+67pp	—
—	Llama 3.2	J2	100	24%	28%	50%	+63pp	—
—	Llama 3.2	K1	100	20%	22%	46%	+53pp	—
—	Llama 3.2	K2	100	23%	26%	64%	+62pp	—
—	Llama 3.2	K3	100	21%	23%	56%	+61pp	—

Table A1. Complete results for all 72 runs. Run IDs marked — are from updated dataset (IDs available in supplementary CSV). All runs $n=100$.

Appendix B: Methodology and Replication Notes

The Agentic API Test Lab is fully open-source and available by request. The pre-registered task rubric, catalog definitions, and condition configurations are provided as supplementary files. All experiments used temperature=0.0 with per-trial seeded catalog shuffling (seed = $42 \times \text{iteration} \bmod 2^{31}$) to ensure reproducibility and eliminate position bias. The results.db SQLite schema is versioned; schema migration is automatic on startup.

Llama 3.2 model note: All Llama runs used llama3.2:latest via Ollama at <http://localhost:11434>. This is the 3B parameter quantized model. Results may differ for the 11B or 90B variants. Researchers replicating with larger Llama variants should expect performance between the 3B and frontier results reported here.